

RESEARCH INTEREST

My research focuses on building trustworthy software and AI systems through automated and empirical analysis. I develop program analysis, fuzzing, testing, benchmarking, measurement, and forensic analysis techniques to identify, understand, and mitigate reliability, security, and privacy problems in real-world systems. My current work spans software engineering, cybersecurity, digital forensics, cloud and infrastructure security, blockchain consensus testing, AI security and safety, and privacy-preserving machine unlearning. For my Ph.D. degree, I studied design problems and vulnerabilities in the Android platform.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Ph.D. in Information Security	Daejeon, KR 2018–2021
– Advisor: Sukyoung Ryu	
– Co-Advisor: Yongdae Kim	
– Thesis: Towards Secure Interoperation in Android: A study of interoperation vulnerabilities in Android	
Korea Advanced Institute of Science and Technology (KAIST) M.S. in Information Security	Daejeon, KR 2012–2014
– Advisor: Yongdae Kim	
– Co-Advisor: Sukyoung Ryu	
– Thesis: Self-Update Vulnerability in Android	
University of Manchester B.S. in Software Engineering	Manchester, UK 2005–2008

PUBLICATIONS

1. Seokhun Jeong, Gyeongmin Dan, Sukyoung Ryu, **Sungjae Hwang**, “SpecTrum: Specification-Guided Differential Fuzzing for Ethereum Consensus Clients”, (ASE 2026)
2. Youngseok Kim, Sungho Lee, **Sungjae Hwang**, “How Safe is Your Screen? Understanding and Detecting Privacy Leaks in Sensitive Activities”, (ISSTA 2026)
3. Seonghyeon Song, Taeyoung Kim, Woojoo Kim, Seojin Park, **Sungjae Hwang**, Hyoungshick Kim, “Ghosts in the Memory: Detecting Unintended Sensitive Data in Android Apps”, (ISSTA 2026)
4. Haeun Eom, Bohyun Suk, **Sungjae Hwang**, “An Empirical Study and Benchmark of Kubernetes Misconfiguration Scanners”, (ISSTA 2026)
5. Jinhee Yoon, **Sungjae Hwang**, “Forensic Analysis of Video Data Deletion and Recovery in Honeywell Surveillance File System”, (DFRWS USA 2026)
6. Dohee Kim, Zhao Hui, Doowon Kim, **Sungjae Hwang**, “Unveiling the Underground Phishing Ecosystem: A 12-Year Longitudinal Study of Deep and Dark Web Forums”, (WWW 2026)
7. Jongchan Hong, Jaewon Kim, **Sungjae Hwang**, “OCPPuzz: Specification-driven Fuzzing of Charging Station Management Systems with Large Language Model”, (FSE 2026)

8. Sanghyeok Park, **Sungjae Hwang**, Simon S. Woo, “From Rules to LLM-Enhanced Templates: A Hybrid ALPG Code Generation System”, (ICSE SEIP 2026)
9. Seohyun Ahn, **Sungjae Hwang**, “Context-Aware Risk Assessment Metric for Kubernetes Admission Webhook Misconfigurations”, (KIISC 2026)
10. Junhyeok Yoo, **Sungjae Hwang**, “A Measurement Study of Scam Activities on NFT Discord”, (CISC-S 2026)
11. Junehyung Kim, **Sungjae Hwang**, “All You Need is Attention: Lightweight Attention-based Data Augmentation for Text Classification”, (EMNLP-Findings 2024)
12. **Sungjae Hwang**, Sungho Lee, Sukyoung Ryu, “An Empirical Study of JVMs’ Behaviors on Erroneous JNI Interoperations”, (TSE 2024)
13. Haeun Eom, Dohee Kim, Sori Lim, Hyungjoon Koo, **Sungjae Hwang**, “R2I: A Relative Readability Metric for Decompiled Code”, (FSE 2024)
14. **Sungjae Hwang**, Haeun Eom, “Challenges in Firmware Re-Hosting for Satellite Systems”, (ANSI 2023)
15. Shinhae Kim, **Sungjae Hwang**, “EtherDiffer: Differential Testing on RPC Services of Ethereum Nodes”, (FSE 2023)
16. Seunghoon Lee, Sukmin Kang, Seungyeon Cho, Hyunwoo Koo, **Sungjae Hwang**, Jinkyu Lee, “RT-Blockchain: Achieving Time-Predictable Transactions”, (RTSS 2023)
17. **Sungjae Hwang**, Sungho Lee, Jihoon Kim, Sukyoung Ryu, “JUSTGen: Effective Test Generation for Unspecified JNI Behaviors on JVMs”, (ICSE 2021)
18. **Sungjae Hwang**, Sukyoung Ryu, “Gap between Theory and Practice : An Empirical Study of Security Patches in Solidity”, *42nd ACM/IEEE International Conference on Software Engineering (ICSE 2020), May 2020*
 - Tool open-source project: github.com/sjmini/icse2020-Solidity
19. **Sungjae Hwang**, Sungho Lee, Sukyoung Ryu, “All about Activity Injection: Threats, Semantics, Detection, and Defense”, *Software: Practice and Experience (SPE), 2020*.
20. Sungho Lee, **Sungjae Hwang**, Sukyoung Ryu, “All about Activity Injection: Threats, Semantics, and Detection”, *32st IEEE/ACM International Conference on Automated Software Engineering (ASE), November 2017*
21. **Sungjae Hwang**, Sungho Lee, Yongdae Kim, Sukyoung Ryu, “Bittersweet ADB: Attacks and Defenses”, *10th ACM ASIA Conference on Computer and Communications Security (ASIACCS), April 2015*
22. **Sungjae Hwang**, “Update Vulnerability in Android”, *M.S. Thesis, KAIST, 2014*
23. Sangyong Choi, **Sungjae Hwang**, “A Technique of Symptoms Analysis over Time for Detection of APT Attacks,”, *International Conference on Smart Convergence Technologies and Applications (SCTA), 2012*

EXPERIENCE

Sungkyunkwan University, Computer Science and Engineering Assistant Professor	Suwon, KR 2021–Present
LG Electronics, Vehicle Solutions Division, Security Team Senior Security Researcher	Seoul, KR 2017–2021
LG Electronics, Mobile Communication Division, Security Team Assistant Security Researcher	Seoul, KR 2015–2017
Korea Internet & Security Agency Assistant Security Researcher	Seoul, KR 2011–2012
ISIS Korea, Inc. Software Engineer	Seoul, KR 2008–2011

TEACHING

- **Instructor** at Sungkyunkwan University
Offensive Security
- **Instructor** at Sungkyunkwan University
Programming Languages
- **Instructor** at Sungkyunkwan University
Web Programming Lab
- **Instructor** at Sungkyunkwan University
Computer Programming for Engineers
- **Instructor** at Sungkyunkwan University
Basis and Practice in Programming

AWARDS AND ACHIEVEMENTS

- | | |
|---|-----------|
| • Ph.D Scholarship from LG Eletronics. | 2018–2022 |
| • Member of South Korea National Chess Team for Summer Universiade | 2013 |
| • 1 st place in Mind Sport Olympiad Chess Tournament | 2010 |
| • Member of South Korea National Chess Team for 38th World Chess Olympiad | 2008 |
| • 1 st place in Mind Sport Olympiad Chess Tournament | 2008 |
| • 3 rd place in Asian Dragon Chess Tournament | 2008 |
| • 1 st place in Mind Sport Olympiad Chess Tournament | 2006 |